

1-3 Propagation of Uncertainty



Source / 來源

- 選修物理(一)課堂講義 (4623R52) 教用版，第1章 不確定度，§ 1-3 不確定度的組合，pp. 10–12
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 1: Measurement
- Young & Freedman, *University Physics with Modern Physics*, Ch 1: Units, Physical Quantities, and Vectors

Examples

Core Ideas — Independent Uncertainty Contributions

When a derived quantity depends on independently measured quantities, their uncertainties propagate into the final result.

For addition or subtraction,

$$z = ax \pm by \implies u(z) = \sqrt{a^2 u(x)^2 + b^2 u(y)^2}$$

For multiplication or division,

$$z = axy \text{ or } z = a\frac{x}{y} \implies \frac{u(z)}{|z|} = \sqrt{\left(\frac{u(x)}{x}\right)^2 + \left(\frac{u(y)}{y}\right)^2}$$

The dimensionless ratio $u(x)/|x|$ is the **relative uncertainty**.

Worked Example — Perimeter, Area, and Average Speed (p. 10; 範例說明)

A rectangular house has length (18.20 ± 0.29) m and width (11.50 ± 0.29) m.

(a) Find its perimeter.

(b) Find its area.

(c) If one lap takes (27.40 ± 0.29) s, find the average speed.

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 Solution

Example 1 — Combination of Uncertainty (*p. 11; 範例 1*)

A sinkhole is measured as 42.20 ± 0.29 m long, 20.30 ± 0.29 m wide, and 21.50 ± 0.29 m deep. Find its volume and uncertainty.

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 Solution

Related Problem 1 — Gravitational Potential Energy (*p. 11; 類題 1*)

A marble has measured mass $M \pm u_1$, height $H \pm u_2$, and local gravitational acceleration $g \pm u_3$. Which expression gives the uncertainty in $U = Mgh$?

(A) $u_1 + u_2 + u_3$

(B) $\frac{u_1 + u_2 + u_3}{3}$

(C) $\sqrt{u_1^2 + u_2^2 + u_3^2}$

(D) $\sqrt{\left(\frac{u_1}{M}\right)^2 + \left(\frac{u_2}{H}\right)^2 + \left(\frac{u_3}{g}\right)^2}$

(E) $MgH\sqrt{\left(\frac{u_1}{M}\right)^2 + \left(\frac{u_2}{H}\right)^2 + \left(\frac{u_3}{g}\right)^2}$

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 Solution

Quick Check

Q1 — Relative Uncertainty of a Cuboid's Volume

A cuboid has dimensions 10.0 cm, 4.5 cm, and 2.0 cm, each with uncertainty 0.5 cm. What is the relative uncertainty of its volume?

(A) 1.50

(B) 0.75

(C) 0.50

(D) 0.42

(E) 0.28

 Solution

Challenge

Photogate Set — Measuring g

[Figure: A U-shaped infrared photogate apparatus measuring a steel ball in free fall.]

Q1 — Complete the Experimental Record

The measured intervals are 0.311, 0.313, 0.311, and 0.310 s. The timer's smallest division is 0.001 s. Complete $\bar{t}$, SD , u_A , u_B , and u .

 Solution

Q2 — Propagate the Result to g

The gate separation is $A \pm a$ m and the measured time is $B \pm b$ s. From $h = \frac{1}{2}gt^2$, express g and its uncertainty in terms of A , a , B , and b using the textbook's independent-factor treatment.

 Solution

Exam Archive

Q1 — Jewelry Scale and Combined Uncertainty (2023, 112 分科)

A balance has a 1 mg smallest division. Five measurements give a combined uncertainty $u_C = 0.35686524$ mg. How should u_C be reported, and how is it combined from u_A and u_B ?

 Solution

Glossary

English term	中文	Definition / Note		
derived quantity	導出量	Quantity calculated from measured inputs		
uncertainty propagation	不確定度的組合／傳遞	Carrying input uncertainties into a result		
absolute uncertainty	絕對不確定度	Uncertainty expressed in the quantity's unit		
relative uncertainty	相對不確定度	$u(x)/$	x	\$
independent measurements	獨立測量	Measurements whose uncertainty contributions are uncorrelated		
quadrature	平方和原則	Root-sum-square combination		
best estimate	最佳估計值	Central value rounded to match the uncertainty		
volume	體積	Product of three lengths		
average speed	平均速率	Distance divided by elapsed time		
gravitational potential energy	重力位能	$U = Mgh$ near Earth's surface		
photogate	光電計時器 ／光電閘	Infrared timing device used to measure motion		